98 — Thesis Synthesis Matrix, Chapter Argument Mapping, and Final Claim Gate

Project Title: Research, Design, and Implementation of an Optimized Gomoku Game Engine with Artificial Intelligence **Academic Module:** Basic Project (■■ Án C■ S■) — Software Engineering & AI Track **Document Standard:** Standard 00 (IMRaD & Evidence Specification for Software Systems) **Control Layer ID:** `SYN-THESIS-98` **Authoritative Repository:** [GitHub – tducn110/caro](https://github.com/tducn110/caro) (commit `cb6fd6f`) **Language:** English **Status:** Complete Synthesis Specification

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1. Purpose & Structural Bridge

While weekly research logs document the chronological process of investigation, the final university thesis dissertation must be organized by argumentative and structural coherence. This synthesis matrix provides the formal bridge mapping empirical findings, architectural decisions, and verified test evidence into the final thesis chapters.

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flowchart LR
    WeeklyLogs["Chronological Weekly Logs\n(Weeks 01 to 10)"] --> EvidenceFilter{"Evidence Control Gate\
    EvidenceFilter --> ThesisChapters["Argumentative Thesis Structure\n(Chapters 1 to 8)"]
```

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2. Thesis Chapter Mapping Matrix

Thesis Chapter	Chapter Focus	Source Research Logs	Canonical Evidence / Artifacts	Key Arguments & Findings
Ch. 1: Introduction & Problem Scope	Problem Motivation, Research Questions	Week 01, Section 00	`RQ-W1-01` to `RQ-W10-04`	Formal combinatorial game definition; intractability of unconstrained search (10^{105}); thesis goals.
Ch. 2: Literature & Theoretical Foundations	Game Theory, Adversarial Search, Pruning	Week 01, Week 05, Week 06	Allis (1994), Knuth & Moore (1975), Shannon (1950)	Proof of first-player win; theoretical branching bounds ($\mathcal{O}(b^{d/2})$); Zobrist hash theory.
Ch. 3: System Architecture & Design	Multi-tier Architecture, Concurrency, WASM	Week 02, Week 03, Section 01	`ADR-ARCH-01`, Migration Decision Gates	Decoupled Web Worker isolation; PixiJS WebGL 60 FPS rendering; zero-copy WASM memory buffer.
Ch. 4: Algorithmic Modeling	Heuristics, Search Trees, Candidate Policies	Week 04, Week 07, Week 08, Week 09	`CFG-EVAL-v1`, `CFG-CAND-v1`, `CFG-ORDER-v1`	7-shape pattern taxonomy; Chebyshev $r=2$ candidate window; dynamic move ordering; Zobrist TT.
Ch. 5: Implementation & Verification	Core Engine, Test Suites, Equivalence Proof	Week 02, Week 06, Section 07	`caro-core-reference` tests (9/9), `10_caro` tests (18/18)	In-place move rollback; search equivalence ($\text{AB} \equiv \text{MM}$); 100% test gate passing.

Ch. 6: Experimental Evaluation	Empirical Scaling, Benchmarks, Tournaments	Week 05, Week 07, Week 08, Week 10	`EXP-W5-01` to `EXP-W10-02`	92.9% node pruning at depth 4; 100% tactical recall; round-robin tournament validation (96% win rate).
Ch. 7: Discussion & Limitations	Engineering Trade-offs, Edge Cases	All Weeks	Horizon Effect analysis, Mobile context loss	Memory limits in WASM; defense multiplier tuning; first-player advantage mitigation.
Ch. 8: Conclusion & Future Scope	Summary of Findings, Future Roadmap	Section 98, Roadmap	Final DoD checklist	Delivery of production-ready engine; prospective deep reinforcement learning extensions.

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3. Final Claim Gate & Quality Assurance Protocol

Before any scientific claim or performance figure is accepted into the final thesis document, it must pass through the six-stage verification gate:

```
flowchart TD
    Claim["Proposed Thesis Claim"] --> G1{"1. Evidence Classification Known?\n([A], [B], [C], [D])"}
    G1 -- "NO" --> Reject["REJECT: Must classify evidence"]
    G1 -- "YES" --> G2{"2. Repository Source Identifiable?\n(Commit / File / Function)"}
    G2 -- "NO" --> Reject
    G2 -- "YES" --> G3{"3. Configuration Version Locked?\n(e.g., CFG-FINAL-v1)"}
    G3 -- "NO" --> Reject
    G3 -- "YES" --> G4{"4. Experiment Reproducible?\n(Controlled variables & dataset)"}
    G4 -- "NO" --> Reject
    G4 -- "YES" --> G5{"5. Boundary Limitations Explicit?\n(Stated conditions & scope)"}
    G5 -- "NO" --> Reject
    G5 -- "YES" --> Accept["ACCEPT: Valid Academic Thesis Claim"]
```

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4. Synthesis Conclusion

By enforcing this bidirectional trace between weekly logs and thesis chapters, every assertion in the thesis is rooted in verifiable empirical evidence, eliminating unverified claims and speculative prose.